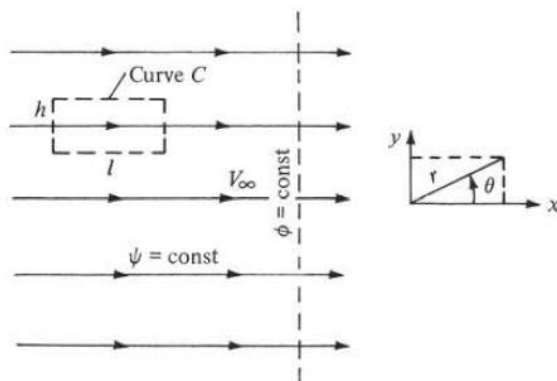
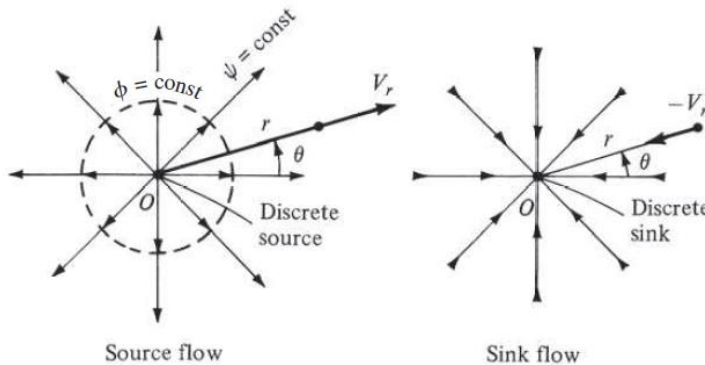


Elementary Flows

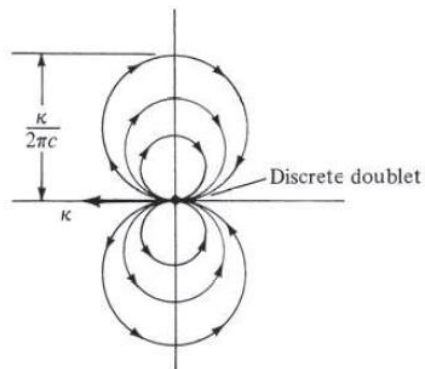
- Building blocks of aerodynamics.
- Small packages of unique flow features that represent building blocks for aerodynamics.
- Incompressible and irrotational (inviscid also).
- Solutions of these elements can be added together to recreate the complex flow field.
- Four main elements: Uniform, Source/Sink, Doublet, and Vortex.
- Polar coordinate system is used.



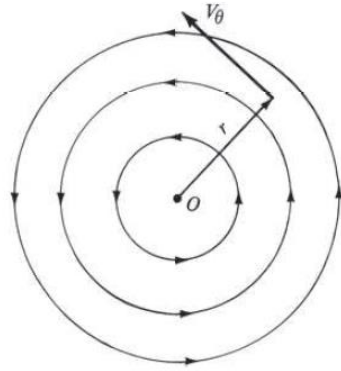
Uniform flow



Source/Sink Flow



Doublet flow



Point vortex

Uniform flow

- Constant velocity field with constant streamwise velocity and zero vertical velocity.
- $u = U_{\infty}$ and $v = 0$
- Physically possible incompressible flow ($\nabla \cdot \vec{V} = 0$) and also irrotational ($\nabla \times \vec{V} = 0$).

Stream function:

By definition:

$$u = \frac{\partial \psi}{\partial y} = U_{\infty} \text{ and } v = -\frac{\partial \psi}{\partial x} = 0$$

Integrating the equations w.r.t y and x we get

$$\begin{aligned} \psi &= U_{\infty} y + f(x) \\ \psi &= \text{constant} + f(y) \end{aligned}$$

Where $f(x)$ is a function of x only and $f(y)$ is a function of y only.

Ψ is the same function, hence $f(y) = U_{\infty} y$ and $f(x) = \text{constant}$. Thus

$$\psi = U_{\infty} y + \text{constant}$$

Note that in a practical aerodynamic problem, the actual value of Ψ is not significant; rather, Ψ is always used to obtain the velocity by differentiation;

Since the derivative of a constant is zero, we can drop the constant from Ψ

$$\psi = U_{\infty} y$$

In the cylindrical coordinate system,

$$\psi = U_{\infty} r \sin \theta$$

Velocity potential:

$$u = \frac{\partial \phi}{\partial x} = U_{\infty} \Rightarrow \phi = U_{\infty} x + f(y)$$

$$v = \frac{\partial \phi}{\partial y} = 0 \Rightarrow \phi = \text{constant} + f(x)$$

Therefore,

$$\phi = U_{\infty} x + \text{constant}$$

In the cylindrical coordinate system,

$$\phi = U_{\infty} r \cos \theta$$

Source/Sink

- Identical flows but opposite in direction.
- These are characterized by streamlines that emit outward or inward along straight streamlines coming from a single point.
- Velocity along each streamline varies inversely with distance from center point o.
- The volumetric flow rate per unit span or depth characterizes the velocity field.
- The radial velocity is expressed as

$$u_r = \frac{\Lambda}{2\pi r} = \frac{\text{Volumetric flow rate per unit span / depth}}{\text{Local circumference}}$$

- The tangential/azimuthal velocity, $u_{\theta} = 0$
- Source flow is a physically possible incompressible flow at every point except the origin.
- Source flow is irrotational at every point.

Stream function:

Integrating the velocity component, we get Ψ

$$u_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{\Lambda}{2\pi r} \Rightarrow \psi = \frac{\Lambda}{2\pi} \theta + f(r)$$

$$u_\theta = -\frac{\partial \psi}{\partial r} = 0 \Rightarrow \psi = \text{constant} + f(\theta)$$

The equation of streamlines

$$\psi = \frac{\Lambda}{2\pi} \theta = \text{constant}$$

Velocity potential:

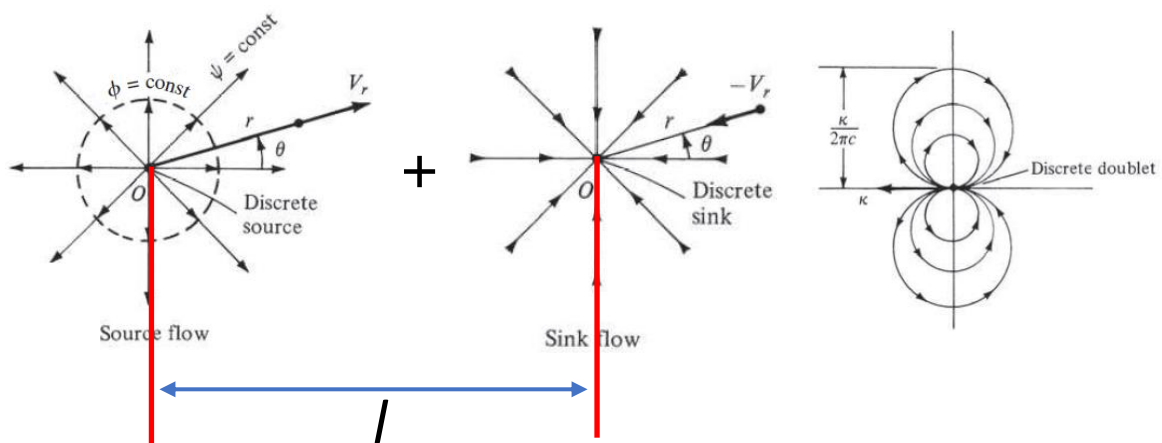
$$u_r = \frac{\partial \phi}{\partial x} = \frac{\Lambda}{2\pi r} \Rightarrow \phi = \frac{\Lambda}{2\pi} \ln r + f(\theta)$$

$$u_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = 0 \Rightarrow \phi = \text{constant} + f(r)$$

Velocity potential: $\phi = \frac{\Lambda}{2\pi} \ln r$

Doublet

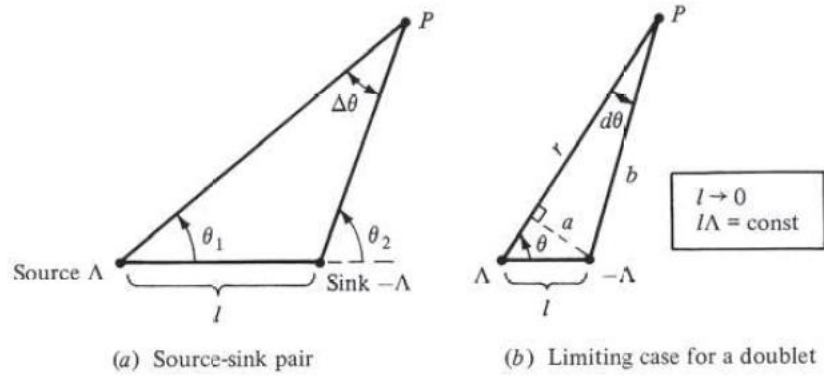
- It is not a unique building block and a combination of source and sink.
- Equal and opposite strength separated by the distance r .



To make a doublet, shrink the distance between source and sink (l) to zero while keeping the parameter $l\Lambda$ (doublet strength) as constant.

Stream function:

$$\psi = \frac{\Lambda}{2\pi} \theta_1 - \frac{\Lambda}{2\pi} \theta_2 = \frac{\Lambda}{2\pi} \Delta\theta$$



let the distance l approach zero while the absolute magnitudes of the strengths of the source and sink increase in such a fashion that the product $l\Lambda$ remains constant. In the limit, as $l \rightarrow 0$ while $l\Lambda$ remains constant, we obtain a special flow pattern defined as a *doublet*. The *strength* of the doublet is denoted by κ and is defined as $\kappa \equiv l\Lambda$.

From trigonometry and small angle approximation,

$$\psi = -\frac{\Lambda}{2\pi} \frac{l \sin \theta}{r} = -\frac{l\Lambda}{2\pi r} \sin \theta = -\frac{\kappa}{2\pi r} \sin \theta$$

From the stream function, velocity potential can be derived.

- Convert the stream function to the velocity field (u, v) and then the velocity field to the velocity potential.

$$\phi = \int \left(\frac{1}{r} \frac{\partial \psi}{\partial \theta} \right) dr = \frac{l\Lambda}{2\pi} \frac{\cos \theta}{r} + f(\theta)$$

$$\phi = \int r \left(-\frac{\partial \psi}{\partial r} \right) d\theta = \frac{l\Lambda}{2\pi} \frac{\cos \theta}{r} + f(r)$$

- Velocity potential: $\phi = \frac{l\Lambda}{2\pi} \frac{\cos \theta}{r} = \frac{\kappa \cos \theta}{2\pi r}$

Vortex

- This is characterized by having a fluid orbiting around some central point.
- The flow field is defined by having no radial velocity component and constant angular velocity.
- Vortex flow is a physically possible incompressible flow ($\nabla \cdot \vec{V} = 0$) at every point.
- Vortex flow is irrotational ($\nabla \times \vec{V} = 0$) at every point except the origin.

- From the definition of vortex flow,

$$u_r = 0$$

$$u_\theta = \frac{\text{constant}}{r} = \frac{C}{r}$$

To evaluate the constant C , take the circulation around a given circular streamline of radius r :

$$\Gamma = -\oint_C \vec{V} \cdot d\vec{s} = -u_\theta (2\pi r)$$

$$u_\theta = -\frac{\Gamma}{2\pi r}$$

Compare the two equations of u_θ , we get

$$C = -\frac{\Gamma}{2\pi}$$

Where Γ is the strength of the vortex flow.

The velocity potential for vortex flow can be obtained as follows:

$$u_r = 0 = \frac{\partial \phi}{\partial r} \Rightarrow \phi = \text{constant} + f(\theta)$$

$$u_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = \frac{\Gamma}{2\pi r} \Rightarrow \phi = \frac{\Gamma}{2\pi} \theta + f(r)$$

$$\phi = \frac{\Gamma}{2\pi} \theta$$

The stream function for vortex flow can be obtained as follows:

$$u_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = 0 \Rightarrow \psi = \text{constant} + f(r)$$

$$u_\theta = -\frac{\partial \psi}{\partial r} = \frac{\Gamma}{2\pi r} \Rightarrow \psi = \frac{\Gamma}{2\pi} \ln r + f(\theta)$$

$$\psi = \frac{\Gamma}{2\pi} \ln r$$

Each elemental flow has a parameter to tune the strength of it:

	Elementary flow	Strength
1.	Uniform flow	U_∞
2.	Source/Sink	Λ or $-\Lambda$
3.	Doublet	$k = \Lambda \lambda$

4.	Vortex	Γ
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Summary:

Uniform flow:

Stream function: $\psi = U_{\infty} r \sin \theta$

Velocity potential: $\phi = U_{\infty} r \cos \theta$

Source/Sink:

Stream function: $\psi = \frac{\Lambda}{2\pi} \theta = \text{constant}$

Velocity potential: $\phi = \frac{\Lambda}{2\pi} \ln r$

Doublet:

Stream function: $\psi = -\frac{\Lambda}{2\pi} \frac{\sin \theta}{r} = -\frac{\Lambda l}{2\pi r} \sin \theta = -\frac{k}{2\pi r} \sin \theta$

Velocity potential: $\phi = \frac{\Lambda}{2\pi} \frac{\cos \theta}{r} = \frac{k \cos \theta}{2\pi r}$

Vortex:

Stream function: $\psi = \frac{\Gamma}{2\pi} \ln r$

Velocity potential: $\phi = \frac{\Gamma}{2\pi} \theta$

Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.